ZOO-514 OBJECTIVES

FINAL TERM- 100%

PRINCIPLES OF ANIMAL LIFE-I

GUESS PAPER 2024

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OBJECTIVES

- Study of animals……**zoology**
- Sperm storage temperature…….**34°c**
- Supporting and binding……. **Connective tissue**
- Role in division……**centrioles**
- MTOC stands for….. **Microtubule Organizing Center**
- Smoth muscle….**spindle shape**
- Smallest unit…….**atom**
- Process of production of an organism from unfertilized egg……..**Parthenogenesis**
- Genetic variability…..**sexual reproduction**
- During ovulation……**LH hormone released**
- Corpus luteum secrets estrogen and progesterone……..**luteal phase**
- Some enzymes require one or more metal ions for their activity called ..... **cofactors**
- ATP is the major currency of ..... **cell**
- In first step acetyl-CoA reacts with ..... to form Citric acid. **oxaloacetate**
- ..... is a type of asexual reproduction in which one organisms splits into two or more organisms. **Fission**
- Around urinary bladder duct deferens enlarges to form ..... **ampulla**
- Head contain ..... **nucleus**
- Egg containing more amount of yolk and less cytoplasm is called ..... **vegetal pole**
- which is not for squamous epithelial tissues..... **Double layered**
- which is not a type of lipid..... **Glycerol**
- The role of flagella..... **Locomotion and sensory**
- smooth muscle..... **Spindle shaped**
- Reptiles egg are.... **Shelled eggs with extra embryonic membranes**

- Protostomes..... **Spiral**
- stages of holo metamorphosis..... **Egg-larve-pupa-adult**
- lentic ecosystem..... **Lackes and ponds**
- coelom is also formed by out pouching..... **Mesoderm**
- chaparral.....**30-75cm**
- world biodiversity day..... **22nd may**
- eutherian mammals..... **20 days to 19 months**
- A ..... Is the parameter used to determine the outcome of an experiment (**dependent variable**)
- Outer layer of cytoplasm is called (**ectoplasm**)
- protostome is an example of **Spiral Cleavage**
- in chappral annual precipitation Wala kch tha ans Uska **30 to 75** tha
- time of low body temperature And decreased metabolism from weeks to months (**hibernation**)
- ek sucrose ka tha K knsa sucrose ni hai ya kisme sucrose ni ota etc kch aesa
- Ek carbohydrates Wala tha k healthy man ko Kitne carbohydrates require otey h etc kch aesa
- Carbohydrates provides .....% energy. **64-65%**
- How many amounts of energy carbohydrates provide per day? **225-325 Grams per day**
- power house of cell Kisko kha jata ek yeh tha _____ **Mitochondria**
- ek lotic ecosystem ka Aya tha k flowing of water Kisko khtey
- desert me konsa animal ni ota kch aesa tha
- Small biomolecules **include** monosaccharides, amino acids and intermediates of different metabolic pathways.
- Glucose is the most common type of **monosaccharides**.
- Cardiac cells are branched, striated, contain single nucleus and connected through intercalated discs.
- Palmitoleic acid---------- **Olive oil**
- Reactions of step 6 to 10 constitute payoff phase**. Total 4 ATPs are produced during this phase**.
- The **central microtubule structure** of cilia and flagella is called axoneme or axial filament. Axoneme has nine pairs of microtubules arranged in a circle and two central tubules. This **is 9+2 pattern**.
- **H +** can come back to matrix through a membrane protein "ATP Synthetase"
- Coelom is also formed by out pouching or splitting of **mesoderm**.
- **Asexual animals:** sexual reproduction is not feasible for animal which rely on asexual means only, especially population with one sex only.
- **Adaptive genes combination dismantle:** In sexual reproduction new genetic combination arise. This will upset the already established adaptive gene combination.
- It is the logarithmic plot between survivors and their age_____ **Survivorship Curve**
- Lentic Ecosystem This is the ecosystem of ***standing water***.
- Parasites living within the body of host are **endoparasites**. E.g., tape worm, liver fluke, plasmodium
- **Density-independent Factors**: Weather conditions • Earth quake • Pollutants • Deforestation • Hurricane

- Hydrogen sulfide, elemental sulfur and sulfide minerals is called **mineralization** and the step at which sulfides and elemental sulfur are converted to sulfates is called **oxidation**.
- Competition between species may be reduced if resources are divided. One part of resources is used by one species and second part by other species. This mechanism is called **niche diversification**.
- Development of many protostome include a top shape larva called **trochophore** larva while some deuterostome have a Kidney-bean shaped larva called **dipleurula**.
- **Pyramid of Numbers** It is the graphical representation of total number of different organisms present at each trophic level. This also shows the population of organisms in a food chain. For example in a grass land the survey was conducted to check the trophic structure. The number of producers was 5.842 x 106 , primary consumers 7.08x105 , secondary consumers: 5.4x104 and tertiary consumers was one.
- It is the representation of total mass of living matter present at each trophic level. Biomass decrease at each ascending trophic level. Biomass is measured **in grams of dry mass per square meter**.
- **Pyramid of Energy** It is the representation of total amount of energy present at each trophic level in an ecosystem. It is always upright. Energy pyramids help to compare the productivity of ecosystems.
- In animals, which organ maintains the volume of water?_**Kidney**
- In a sequence of hierarchy in which the female comes first, followed by the male, it is called: **Matrilineal/ protogynous**
- Amino acids are linked through: **Peptide bond**
- The main function of animals is to maintain: **Homeostasis**
- The process by which organic sulfur is converted into inorganic form i.e., hydrogen sulfide, elemental sulfur and sulfide minerals is called **mineralization**
- The step at which sulfides and elemental sulfur are converted to sulfates is called **oxidation**
- A **dependent variable** is the parameter used to determine the outcome of an experiment
- monosaccride form together... **Polysaccharides**
- These are the animals which retain eggs into the body but the young one gets all its food from yolk of egg. …. **Ovoviparous**
- Ammonia is formed during deamination of?_**Amino Acid**
- Who enzyme is essential for vertebrate skeleton and teeth?_ **Phosphorus**
- Outer layer of cytoplasm is called?__ **Ectoplasm**
- Name the simple sugar that are not further hydrolyzed? __ **Monosaccharides**
- Which is essential component of DNA, RNA, phospholipids?_ **Phosphorus**
- In mating process male introduces sperms into female body, sperms are motile they approach to female gametes within the body of female. Many sperms are packed into a ball like structure called **spermatophore**
- During first five months of development many small follicles are formed in the ovary of female fetus. These are called **primordial follicles.**

- In human the entire process of **spermatogenesis** and its transportation takes about **3 months**
- The total gestation period in human is **280 days**. This duration is divided into 3 months periods called **trimesters**.
- **Heart beat** can be heard by the end of **fourth month (18 Week).**
- Write the name of procedure that occure aftr the 4 month of pregnancy?__ **Heart beat**
- Write the name of reproduction that produce new offspring?__ **Sexual Reproduction**
- **Sexual Reproduction** It the types of reproduction in which new offspring are produced by the fusion of gametes
- Protostomes is an example of?__ **Spiral Cleavage**
- Biological science that addresses the questions of proximate causes is called **experimental science.** Which is break into two and more picese of fragments?__ **Fragmentation**
- **Fragmentation** is the process of asexual reproduction in which an animal breaks into two or more pieces
- The process in which one organism divides into two organisms is called **binary fission**
- **Ecology**_It is the branch of biology that deals with the relationship of organisms to one another and with their physical environment
- Organisms require energy in one form or the other.___ **Energy Flow**
- **Symbiosis** is a close and long term association between two different organisms
- Organisms are entirely dependent on symbiosis are obligate. E.g., **Fungus of lichen**
- **Viviparous** __These are the animals in which eggs develop into the body and the young one gets its nourishment from mother.
- Eggs with moderate amount of yolk are called **mesolecithal** eggs.
- e.g., lung fish and amphibians. While eggs with large amount of yolk are called **macrolecithal** eggs. e.g., reptiles and birds.
- Seminiferous tubule join a network of tubules called **rete testis**.
- cleavage occur in embryo of nematodes___ **Holoblastic Cleavage**
- **Holoblastic Cleavage:** The pattern of cleavage in which egg is completely divided into two cells
- “Two monosaccharides combine together to form a **disaccharide**
- Melting point of Palmitic acid... **62.9**
- **Histology**: Study of organisms at tissue level is called histology.

- **Entomology**: Entomology is a branch of zoology which deals with the study of insects.
- **Renewable resources** are those resources which are constantly available
- **Animal diversity** is the number of different types of animals living on this earth.
- An **independent variable** is parameter being tested in the experiment.
- A **dependent variable** is the parameter used to determine the outcome of an experiment while a **control variable** is that which can influence the results so it is kept constant.
- In chaparral annual precipitation.. **30-75cm**
- **Atmosphere** is the gaseous content of biosphere
- The final community during an ecological succession is called **climax community**.
- **Camouflage** It is the phenomenon in which the colour pattern of animal helps to hide the animal from others
- **Hibernation**: Time of low body temperature and decreased metabolism from weeks to months
- Eggs containing negligible amount of yolk are called **alecithal egg**. e.g., placental mammals. Amphibians have...eggs__ **mesolecithal eggs**
- **Radial cleave** is the characteristic of deuterostomes.
- **Meroblastic Cleavage:** The pattern of cleavage in which egg is not completely divided into two cells
- **Holoblastic Cleavage:** The pattern of cleavage in which egg is completely divided into two cells.
- **Parasitism**: It is a relationship in which on organism (parasite) live in or on other organism (host). Parasite gets benefit while host is harmed.
- The cells developing embryo and placenta release **human chorionic gonadotropin (hCG)**
- The tiny chorionic villi sent out from **blastocyst** form placenta
- Human have placenta formed form **chorion and allantois**
- Which is water soluble _ **Globular Protein**
- RNA molecules in nature work as enzymes are called **ribozymes**.
- The non-protein organic molecules require for enzyme activity is called **coenzyme**. e.g., vitamins
- An enzyme without its cofactor or coenzyme is called **apoenzyme**. Apoenzyme is catalytically inactive.
- An enzyme with its attached cofactor or coenzyme is called **holoenzyme**

- **Coniferous forest** are dominated by cone bearing, needle shape evergreen plants called **conifers**
- **Coniferous forests** are present in Chitral, Swat, Dir, Abbot Abad, Hilly Areas of Balochistan, Murree and Azad Kashmir
- Which is essential component of DNA, RNA , phospholipids , vertebrates__**phosphorus**
- Biomass is _
- At an average there is **10%** biomass conversion efficiency at each trophic level.
- It is the efficiency by which animal at a trophic level convert food into new **biomass**
- **Pyramid of Biomass:** It is the representation of total mass of living matter present at each trophic level.
- The amount of energy stored in the form of biomass is called **net productivity**.
- **Logistic Population Growth:** S-shaped curve: At carrying capacity population growth levels off
- **Exponential Growth**: Population growth in which number of individuals increase by a constant multiple in each generation
- **Polysaccharides:** "When many monosaccharides join together through glycosidic bod, they form a polysaccharide"
- Outer layer of cytoplasm is called _ **Ectoplasm**
- Parthenogenesis is a process of production of an organism from an **unfertilized egg**.
- ***Ectosymbiosis***: It is a symbiosis in which symbionts live on the body surface of the host or in the digestive tact. E.g., Lice, barnacles and cleaner fish
- **Endosymbiosis**: It is the relationship in which one symbiont live within the tissue of other. E.g., nitrogen fixing bacteria in root nodules, algae inside the corals
- How much NADH are yields during glycolysis___**Two**
- Marine water typically contain...Percent salt__**97%**
- Fresh water contains salt concentration about ___**3%**
- Less than **1%** of water is available to human.
- In temperate deciduous forest temperature varies from__**-30oC to 30oC.**
- Following stages of premature birth__ **Third Trimester**
- **Neurulation**: The process of forming neural tube is called neurulation.
- **Phagocytosis**: It is the uptake of solid materials by the cell.
- **Pinocytosis**: It is the uptake of liquid from extracellular environment.
- One gram of proteins give approximately 4kcal energy

- Shelled eggs with extraembryonic layers in reptiles__ **amniotic eggs**
- **Reptiles** are true terrestrial vertebrates. They have shelled eggs with extraembryonic membranes.
- **The Follicular Phase**: Developing follicle produces estrogen
- **histology** is study at tissue level
- Average precipitation is very low approximately less than
- 25cm/year in **Tundra Ecosystem**
- Kelps and algae are plants in **Neritic Ecosyste**
- **Columnar epithelial tissue** are single layer of elongated cells
- Beekeeping for the production of honey and wax is called **apiculture** while rearing of silkworm for production of silk is called **sericulture**.
- **Fundamental Properties of Life:** "The ability to reproduce is
- Life."
- Coniferous forest have annual temperature **10°C**
- **Taxonomy**: Brach of zoology that deals with naming and classification
- Egg takes **5 days** to reach uterus
- Amphibian's are_ **Viviparous**
- Amphibians mostly have **external fertilization** in water in few internal.
- Formation of sperm is called...
- The process of sperm production is called **spermatogenesis** □

  Gametogenesis is the process of formation of gametes.
- Which of the following hormones is secreted from pituitary gland__

  Prolactin
- 1 gram protein has....**4 kcal energy**
- Kelps and algae are **plants**
- About **78%** of precipitation occurs over oceans.
- **DESERT ECOSYSTEM:** It is a region which receives annual precipitation less than **25cm**
- Beekeeping is....**apiculture**
- **Sexual reproduction** is consider as more advantageous than asexual reproduction.
- **Reproduction** is the process of producing new individuals
- Average heart beat for human is **72 times/min** or **100,000 times/day**. In the life of 70 years heart beats for **about 2.5 billion times**
- The recommended amount of energy for our daily life is **30%**
- Those sequential hermaphrodites which are female in first phase of life and male in second phase of life called **protogynous** (femle first) e.g., wrasses.
- Those animals which are male in first phase of life and female in second phase of life are called **protandrous** e.g., clone fish.
- Potential life span of American robin is **11 years**, however average life **is 1.4 year**

- It is the first community established in an area devoid of life is called **pioneer community**
- **Water** is essential for life
- Parasites living on the body surface of host are **ectoparasites**.
- E.g., lice, leeches
- Parasites living within the body of host are **endoparasites**. E.g., tape worm, liver fluke, plasmodium.
- Food chain is important To understand the problems of
- **biomagnification**.
- One part of resources is used by one species and second part by other species. This mechanism is called **niche diversification**.
- Some bacteria convert nitrates back to nitrogen is called **denitrification**.
- Current documented biodiversity of earth is more than **1.6 million**
- The total Animal diversity of life documented is about **1.74 million**
- More than **1.5 million** species of animals have been documented.
- **Ecosystem:** It is the study of all population in a community and their physical environment
- **Hemimetabolous** metamorphosis (Incomplete metamorphosis) is the mode of development of certain insects like aphid, grasshopper and cockroach.
- Arctic tundra is present in northern **hemisphere**
- **Hibernation**: Time of low body temperature and decreased metabolism from weeks to months
- **Coexistence**: It is the phenomenon in which competing species live together Pyrimed....
- **Carbon** is the basic element of all organic molecules.
- Melting point **increase** with increase in number of carbon atoms
- **Sprouters** are dormant meristem, storage and intact root help to regenerate quickly while **seeders** are long lived seeds, germination induced by fire.
- **Lotic Ecosystem:** It is the aquatic ecosystem of flowing water. It includes brooks, streams and rivers
- **Lentic Ecosystem:** This is the ecosystem of standing water.
- Lentic ecosystem include lakes and ponds.
- Study of complete structure of organisms and body parts (**anatomy**)
- conclude something from general information (**deductive reasoning**)
- A **deductive reasoning** hypothesis is that conclude something from general information while
- **Inductive reasoning** hypothesis given some general conclusion from very specific information.
- carbohydrates that cannot be hydrolyzed (**monosaccharides**)

- ___is involved in cell recognition and determination of cell behavior (**Glycocalyx**)
- ____is the process that Force small molecules across selective
- permeable membrane using hydrostatic pressure (**filtration**) Double membrane organelle involved in energy production (**mitochondria**)
- Double membrane structure that separate nucleus from cytoplasm (**nuclear membrane**)
- Tissue present in lungs air sacs lining of heart and blood vessels (**simple squamous epithelial tissue**)
- The reaction which absorb energy (**androgenic reaction**)
- ___NADH are used during glycolysis (**2**)
- Longest gestation period is in (***elephant***)
- Which complete after fertilization (**meiosis 2**)
- Deamination of what produces ammonia (**amino acid**)
- Tropical rainforest have annual rainfall more then (**200cm**)
- The rigid material on the surface of earth (**lithosphere**)
- Group of different species (**community**)
- Phenomenon in which ecologically related species exert strong selective influence on other (**coevolution**)
- **Physiology**: The study of the function of organisms and their body parts.
- Tropical rainforest have **50% of plants** and animals diversity with just **6% land**.
- The process of sperms production is called **spermatogenesis** while the process of eggs production is called **oogenesis**.
- In eutherian mammals embryo is implanted deeply, fetus get nutrition through placenta, gestation period **20 days to 19** months.
- Average size of Ostrich egg is **15 x 13** cm
- **Biotic Potential:** The capacity of a population to increase maximally is called biotic potential
- **Temperature**: It is the measure of heat present in a body **Ectoderm**: It is the outer most layer of cells formed during gastrulation.
- **Endoderm** is the inner most primary germ layer formed during gastrulation
- **Mesoderm** is one of the primary germ layer formed during gastrulation. It is present between ectoderm and endoderm.
- In protostomes cleavage is **spiral** while in deuterostomes cleavage is **radial**.
- **Cleavage** refers to cell divisions that occur during embryonic development.
- Initially all the cells of embryo divide at the same time this is called **synchrony**.
- As development proceeds, all cells don't divide simultaneously. This is called **asynchrony**.

- **Holoblastic Cleavage**: The pattern of cleavage in which egg is completely divided into two cells.
- **Meroblastic Cleavage**: The pattern of cleavage in which egg is not completely divided into two cells. This is due to large amount of yolk
- **Spiral Cleavage**: It is pattern in which cells of animal pole are not exactly laying above the vegetal pole cells. Cleavage takes place at oblique angle forming a spiral pattern. Example: protostomes cleavage
- **Radial Cleavage**:It is pattern in which cells of animal pole are exactly laying above the vegetal pole cells. Upper tier of cells is exactly on the lower tier of cells. Example: deurostomes cleavage**.**
- **Kingdom** is the highest rank of classification.
- **Placental mammals** egg is **smallest** in animals kingdom.
- **Mammalian** cleavage is **slowest** in animal kingdom
- Protostomes belong to group **protostomia** of animal kingdom.
- Deuterstomes belong to group **deuterostomia** of animal kingdom.
- **Anatomy**: It is the study of the structure of complete organisms and their body parts.
- **Physiology**: The study of the function of organisms and their body parts.
- **Cell biology (Cytology**): Study of the structure and function of cell and its components.
- **Molecular Biology:** Molecular biology is the study of structure and function of the macromolecules (e.g. proteins and nucleic acids, Lipids) essential to life.
- **Histology:** Study of organisms at tissue level is called histology.
- **Embryology:** Study of development of embryo from fertilized egg.
- **Genetics:** Study of the mechanisms of transmission of traits from parents to offspring.
- Simple form of carbohydrates (**Monosaccharides**)
- Study of the structure of complete organisms and their body part (**Anatomy**)
- Information about Environmental problem (**Data**)
- **Observation** is the collection of information about a particular problem
- The information obtained from observations is called **data**
- Example of Tertiary Structure (**Amylase**)
- Bee keeping is called (**apiculture**)
- Precipitation of Chaparral (**30-75cm)**
- In Protostomes coelom is formed by splitting of mesoderm. This pattern is called (**schizocoelous**)
- Biggest of all levels of Ecological hierarchy (**Biosphere**)
- Outer most layer of cells formed during gastrulation (**Ectoderm**)

- Ectodermal cells migrate towards blastopore and spread over entire embryo. This is called (**epiboly**)